

A review of strategies, challenges, and ethical implications of machine learning in smart manufacturing

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ABSTRACT

Manufacturing organizations continuously need to innovative production strategies and advance their machinery to adapt to evolving business objectives. Machine learning and data mining are now essential techniques for solving various complex manufacturing problems promptly and intelligently. This article reviews recent research from multiple sectors that have employed machine learning to develop intelligent manufacturing processes, while highlighting key challenges and areas that have been partly overlooked. Over the last two decades, scholars have developed numerous AI-based algorithms and approaches to improve manufacturing processes outputs, with scheduling, monitoring, quality, and fault detection being among the main focus areas. The review categorizes smart manufacturing problems into clustering, classification, and regression tasks, and discusses the underlying performance metrics associated with each category. Additionally, the study tackles ethical issues by discussing such important considerations as data privacy, transparency, and fairness in industrial machine-learning implementations. Finally, it emphasizes that many users remain concerned about compliance with global data protection legislations and the need to build trust in autonomous decision-making systems.

1. Introduction

Machine learning (ML) has already brought a paradigm shift in manufacturing, enabling intelligent decision-making tools, automation, and predictive analytics [1,2]. ML has enhanced efficiency and adaptability in industrial environments, from predictive maintenance to quality control. However, existing ML applications in manufacturing also face significant challenges, including real-time adaptability, handling noisy and heterogeneous data, and lack of scalability across different production settings [3,4]. Besides, the integration of ML in manufacturing raises pressing ethical and regulatory concerns. Data privacy, transparency, and fairness are among considerations for responsible AI adoptions. It is of utmost importance that operational and customer data be protected from misuse, bearing in mind the obligations posed by global data protection regulations like the General Data Protection Regulation (GDPR) [5]. Equally important, trust in ML-based autonomous systems can only be built if transparency in decision-making is ensured and the potential bias in algorithms is mitigated. Creating a design and deployment mechanism for embedding these principles within the ML models (e.g., see [6–17]) would allow manufacturers to create a more ethical and sustainable smart manufacturing landscape.

1.1. Proper data acquisition as a prerequisite to successful ML implementation in manufacturing

Data acquisition systems play a preliminary, but key role in any smart manufacturing implementation, by virtue of enabling automated data collection and its electronic transmission from various units of the facility [18]. Raw materials, environmental conditions (such as humidity and temperature), sensor data, equipment malfunctions and maintenance, and product quality can all influence the end-product quality and cost. Such information can also be linked to various other aspects of a manufacturing system, such as product design, machines capabilities, the production line, the human resources involved, among others.

With IT technological advancements, many manufacturing industries now generate vast amounts of raw data daily. This increase in data has also led to growing academic interest in ML applications. Since about two decades ago, ML applications have been emerged to address a range of manufacturing issues, including intelligent systems to aid decision-making, the scheduling of multiple production lines, and plans for machine maintenance [14]. Studies have extensively examined the impact of the ML on the production efficiency and automation

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Nomenclature

ML	Machine Learning
R^2	Coefficient of Determination
RMSE	Root mean square error
KPI	Key Performance Indicator
DT	Decision Tree
NN	Neural Network
SVM	Support Vector Machines
KNN	k-nearest neighborhood
NB	Naive Bias
RF	Random Forest
SMOTE	Synthetic Minority Oversampling Technique
LOF	Local Outlier Factor
ARM	Association Rule Mining
MinSup	Minimum Support
MinConf	Minimum Confidence
SPM	Sequential pattern mining
FCloSM	Frequent Closed Sequence Mining
DDBS	Distributed Database System
HDFS	Hadoop Distributed File System
DL	Deep Learning
RL	Reinforcement Learning
DNN	Deep Neural Network
GA	Genetic Algorithm
CNN	Convolutional Neural Network
DRNN	Deep Recurrent Neural Network
DBM	Deep Boltzmann Machine
RBM	Restricted Boltzmann Machine
DBN	Deep Belief Network
TM	Text Mining
DM	Data Mining
CMIM	Conditional Mutual Information Maximization
MES	Manufacturing Execution Systems
IIoT	Industrial Internet of Things
COLT	Computational Learning Theory
CPS	Cyber-Physical Systems
SMOTE	Synthetic Minority Oversampling
XAI	Explainable Artificial Intelligence
DBSCAN	Density-Based Spatial Clustering of Applications with Noise
DQN	Deep Q-Networks
PG	Policy Gradient
AC	Actor-Critic
MARL	Multi-Agent Reinforcement Learning

[19,20]. Case studies, however, have mostly employed conventional statistical models or basic ML techniques, limited in handling complex, real-time manufacturing environments. Previous reviews in the field, have either focused narrowly on specific subfields such as electronics, additive manufacturing, or semiconductors or lacked a comprehensive framework that integrates diverse ML strategies and concurrent (and sometimes interlinked) challenges [21], such as data heterogeneity, imbalance, and real-time adaptability [15,22].

Deep learning models have been widely used in defect detection and image-based quality control [23–25], yet their lack of interpretability and high computational demands hinder industrial adoption. Similarly, supervised learning approaches, while effective, require large volumes of labeled data, which is often impractical in manufacturing environments. Some studies assume clean and structured datasets, whereas

real-world manufacturing data is highly imbalanced, noisy, and subject to uncertainties.

Upon automated data collection and its processing, manufacturing Execution Systems (MES) and the Industrial Internet of Things (IIoT) can provide a solid basis for data-driven decision-making in smart manufacturing environments. The integration of these technologies develops an all-way connection between shop-floor operations and higher-level enterprise systems [26,27]. MES is an example of a centralized software platform for managing, monitoring, and synchronizing manufacturing processes across production lines. Within the ERP system, MES closes the loop on shop-floor automation by enabling real-time data exchange among the machines, sensors, and human operators. MES, in compiling important metrics about work order execution, production scheduling, resource usage, and product quality, serves as a valuable dataset for several ML models [26].

IIoT implements smart sensors, actuators, and connected devices over the manufacturing facility floor, thus extending the capabilities of traditional automation systems [20]. These sensors continuously harvest data on machine conditions, environmental factors (temperature, humidity, vibration), energy consumption, and operator performance. IIoT enables real-time monitoring, predictive maintenance, and process optimization by aggregating this data into cloud-based or edge-computing platforms. Integrating IIoT with MES provides a structured data flow, contextualized for better ML applications concerning manufacturing. The convergence of MES and IIoT with ML allows manufacturers to move beyond reactive process management toward predictive and prescriptive analytics, enabling intelligent decision-making for operational improvements [28].

1.2. Objective and contribution

To further dissect the above overarching advancements and challenges in the field, this review paper aims at answering ***What are prominent multi-sectorial examples and key components of ML-driven solutions in manufacturing environments, along with the reported implementation barriers, ethical concerns, and potential future directions?***

Unlike previous similar investigations focusing on isolated ML techniques or specific industrial subfields [29–31], this review attempts to synthesize more general insights from various ML paradigms and application sectors, and thereby present a holistic perspective on their impact as well as limitations. Specifically, the study offers the following contributions:

1. It provides a structured analysis of the current smart manufacturing solutions, considering various task types and implementation algorithms (supervised, unsupervised, semi-supervised, and reinforcement learning).
2. It identifies significant issues that need to be addressed in the existing literature, including those impacted by new technological developments.
3. It presents a bibliometrics related to the studies conducted between 2000 and 2024 from different perspectives, including the subjects of the studies and the types of learning employed.
4. It highlights the advantages and potential obstacles associated with utilizing ML techniques in manufacturing, along with recommendation for addressing them.
5. Finally, it explores potential future research avenues that could lead to the discovery of new applications of ML in manufacturing.

Within the ML part, we specifically cover the following notions:

- Review the common ML strategies currently used in diverse manufacturing sectors.
- Explore the potential applications of deep learning (DL) models for solving complex production issues.

- Examine some of the most widely used standard industrial datasets and prominent *open-source* data processing tools that can be applied in the industrial settings.
- Discuss select criteria commonly used to evaluate the performance of ML models for the manufacturing sector.
- Highlight the advantages of using ML techniques in real-world scenarios, including increased efficiency, reduced downtime, and improved product quality.
- Identify some of the largest current challenges facing the widespread adoption of ML in manufacturing.
- Highlight the need to consider ethics and data privacy in further casting ML applications. This would include better compliance with data protection statutes, transparency, and addressing algorithmic bias—primarily aimed to yet-to-fully-build trust in autonomous decision-making systems.

1.3. A categorization of main ML implementation barriers

While ML continues to significantly transform several manufacturing processes, challenges (and hence research gaps) still hinder its broader and effective implementation. The following is a topical classification of these underlying barriers, which will be then reviewed in more details in subsequent sub-sections.

1. **Data Quality and Processing:** One of the most important limitations in applying ML models in manufacturing includes quality, consistency, and availability. The heterogeneous and noisy data in the manufacturing environment emanate from several sources, such as sensor data, quality control logs, and machine parameters. Most previous research, however, assumes a level of cleanliness and structure in the data rarely realistic under real industrial conditions. This calls for further research into data pre-processing techniques to handle this variability, including developing advanced noise reduction, imputation methods in the event of missing data, and efficient handling of imbalanced datasets.
2. **Real-Time System Integration:** Data should continually be integrated into a smart system, allowing it to make predictions and act on them. Predictive maintenance and process optimization require an integrative capability to process and act upon real-time data. From this perspective, there are substantial computational demands besides model updates, as real-time data continuously flows in. It is desired to arrive at lightweight, adaptive ML algorithms that learn and adapt on the fly with low computational overhead. This gap might further be narrowed down when better integration frameworks connect ML models with the manufacturing control systems, thus giving seamless feedback loops from data collection to analytics and operational adjustments.
3. **Ethical and Transparency Considerations in Autonomous Decision-Making:** As manufacturing enters an era with a growing reliance on autonomous decision-making through ML, concerns regarding transparency and accountability arise when ML-driven decisions impact quality control, safety, or human labor. Most ML models, especially deep learning approaches, are not very transparent, complicating accountability and stakeholder trust. Thus, this gap needs careful efforts related to interpretable ML models in manufacturing, where decision-making processes should be more transparent and traceable. Such transparency is important in maintaining ethical and regulatory standards, most notably in those manufacturing environments where safety and quality prevail.
4. **Limited Application of Advanced ML Techniques:** Although some advanced ML techniques such as reinforcement learning, meta learning, and semi-supervised learning hold great potential for dynamic and complex environments typical in manufacturing,

their applications in the field remain rather limited. For instance, reinforcement learning can optimize production by learning from actions or even the consequences of such actions. Still, implementation has been constrained by the nature of manufacturing processes, which are often complex and non-predictive. Adapting RL and semi-supervised approaches to concrete manufacturing challenges, such as optimization of production schedules or handling rare but critical fault conditions, may provide useful advantages from research.

5. **Scalability and Adaptability:** Most ML models developed for manufacturing lack scalability and adaptability because they were only designed for specific tasks or datasets. This limits their generalization in various other manufacturing scenarios or across different production lines. Efforts must, therefore, be channeled toward developing modular, scalable ML frameworks that can adapt to changes in manufacturing conditions and be easily transferred from one application to another. This is a research area in which e.g. transfer learning and model reutilization would save costs and reduce time by retraining for new applications, improving the capability of scalability.

Building on the above overarching challenges, the rest of this review has been structured as follows. Section 2 provides a background information on the interrelationships between various application sectors that contributed to research in smart manufacturing. It also presents a bibliometrics of studies conducted in the preceding two decades, that have received widespread citations for the reliability of their findings. This section also classifies the objectives of such manufacturing studies and exemplifies the factors analyzed in the research relevant to manufacturing. Additionally, it addresses prominent manufacturing-related benchmark datasets. In the following section (Section 3), the paper details the sequential procedures essential for extracting insights within smart manufacturing settings. It delineates the required actions to be taken and clarifies the appropriate transmission of connected data across various stages. Section 4 classifies ML applications into supervised, unsupervised, deep learning, semi-supervised learning and reinforcement learning, and highlights their key implementations that are widely acknowledged. Section 5 highlights decision implications for adopting ML approaches in manufacturing. Section 6 addresses ethical considerations, emphasizing data privacy, compliance with regulations, and the need for transparency and fairness in ML applications to build trust. Section 7 provides a closing discussion and future perspective followed by Section 8 outlining summary of main findings.

2. State of the art and bibliometric analysis

Traditional mathematical and statistical methods are inadequate for depicting in-depth connections between different data types and making accurate predictions for new situations. This implies that a conventional production sector must enhance its models to maximize its capabilities. However, the integration of ML in manufacturing is grounded in several key theoretical frameworks that support its effectiveness during application. One such framework is Computational Learning Theory (COLT), which provides the foundation for understanding how ML models generalize from data, making it essential for defect detection, predictive maintenance, and production optimization. Cyber-Physical Systems (CPS) under Industry 4.0 have revolutionized modern manufacturing by integrating ML-driven decision-making, enabling real-time monitoring, anomaly detection, and self-adaptive control systems [32,33]. Another key framework is Big Data Analytics, which allows ML models to process large, heterogeneous datasets using advanced statistical inference and feature selection techniques. These frameworks enable ML to address complex manufacturing challenges, such as predictive maintenance, process optimization, and adaptive quality [34,35]. Machine Learning is considered a valuable asset in the business sectors because it helps to gain insights and offers an

edge over the competition through intelligent data processing. However, workforce with strong backgrounds in manufacturing and ML is still rare. Therefore, the most common approach to utilizing ML in production is through a collaborative effort between data scientists and manufacturers. Data gathered from manufacturing plants by data technology systems provides input for ML models. Data analytics is used to identify essential features and structures, followed by ML to develop and train effective models for manufacturing system data [36,37]. Data analytics has proven to be highly beneficial for decision-making in the manufacturing industry. Other manufacturing tasks, such as process mining, sales forecasting, and layout planning, can also be improved with the help of ML capabilities. A wide variety of other manufacturing tasks, including material characterization, product design, time/cost prediction, and energy consumption prediction, can all benefit from the application of ML [38–40].

Past studies have examined the uses of ML in manufacturing, focusing on both advantages and challenges [1,2]. Techniques from deep learning, including Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), are accurate in diagnosing machine failures and forecasting breakdowns [41]. Such models, however, generally demand a large amount of labeled data, not to mention the issues of interpretability and computation cost. Reinforcement Learning has been implemented in optimizing production scheduling, dynamically adapting machine parameters, and reducing manufacturing downtime. RL has shown promise in increasing efficiency; however, its implementation is still very limited due to data requirements and the complexities of training models in dynamic environments. Supervised learning models like Support Vector Machines (SVMs), Random Forests (RF), and Decision Trees (DT) have seen great application in real-time defect detection and quality assessment [6,42]. Nonetheless, as stated earlier, most studies assume clean and balanced datasets, while, in reality, manufacturing data is often noisy, imbalanced, and prone to uncertainty. Such challenges have prompted specific techniques, such as Synthetic Minority Oversampling (SMOTE) and anomaly detection methods. Still, their scalability in real-world scenarios in a vast manufacturing system remains questioned [43].

In our bibliometric analysis, to ensure relevance and quality, eligibility criteria included peer-reviewed articles published between 2000 and 2024, written in English, and directly addressing ML approaches such as supervised, unsupervised, and reinforcement learning—used in manufacturing-related contexts like predictive maintenance, quality control, process optimization, energy efficiency, and real-time monitoring. A total of 339 relevant articles were selected. Of these, 214 articles were retrieved from Scopus and 125 from Web of Science (WoS). These databases were chosen due to their extensive coverage of multidisciplinary, high-impact research and their reliability in indexing cutting-edge developments in engineering, computer science, and industrial technologies. Scopus and WoS offer advanced filtering, citation tracking, and thematic categorization features, which helped ensure a balanced and inclusive selection of literature.

Most of the current literature focuses on isolated ML methodologies rather than a comprehensive methodology integrating several paradigms of ML; namely, supervised, unsupervised, and reinforcement learning. Besides, there are significant challenges to real-time data adaptation, scalability, and ethics of AI-based manufacturing. To give a general overview of manufacturing ML applications, we considered a wide gamut of literature from the top academic sources relevant to the particular topic, ensuring several angles of view are represented. Examples in the literature were predictive maintenance, process optimization, and quality control, highlighting the evolving role of ML in manufacturing.

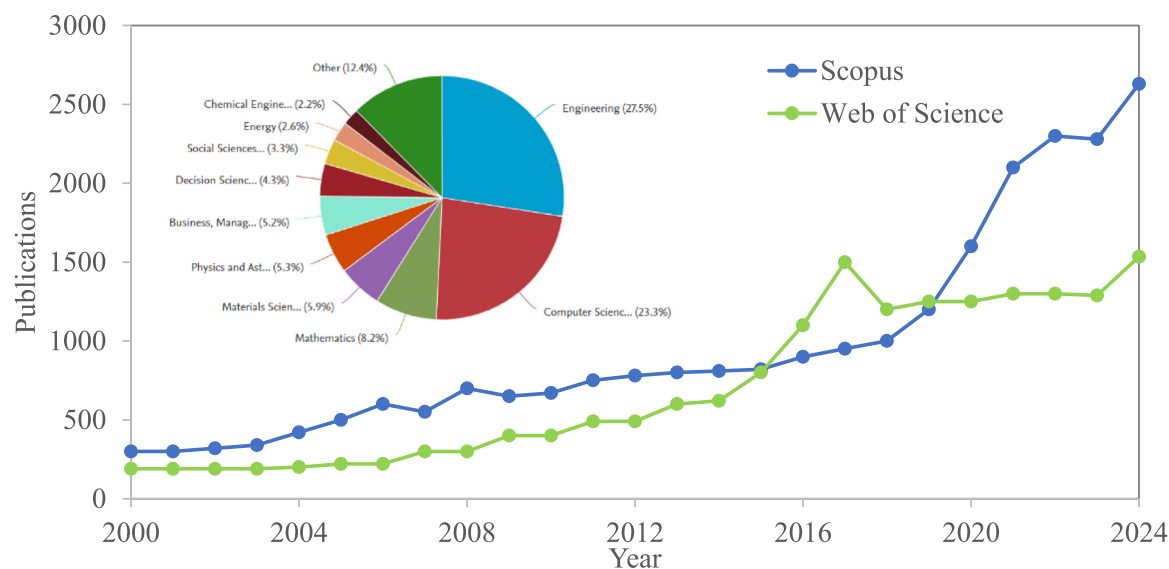
The analysis upward trend in ML-related research, as depicted in Fig. 1a, shows growing interest in utilizing AI-powered approaches to solve manufacturing challenges. This burgeoning has happened not just through technology development, but also because of the demands from the industry, like the necessity of increasing efficiencies and

reducing downtime, along with the need to improve products quality. A clustering methodology was next implemented in our analysis, using VOSviewer, to identify key themes more comprehensively. This tool enabled the systematic categorization and visualization of the research landscape, focusing on ML applications in manufacturing. Keywords such as “predictive maintenance”, “process optimization”, “quality control”, and “cyber-physical systems” were employed to refine the search criteria and ensure relevance to the domain. The clustering process utilized bibliometric analysis to group articles based on the co-occurrence of keywords, citation relationships, and thematic focus areas. VOSviewer generated a network map (Fig. 1b) that provided insights into the connections between articles and their underlying themes. The resulting clusters represent distinct research domains, such as predictive maintenance, quality control, and process optimization, while also revealing emerging trends in areas like energy-efficient manufacturing and supply chain optimization. Each cluster was analyzed in detail, identifying representative articles that exemplify the scope and significance of the theme.

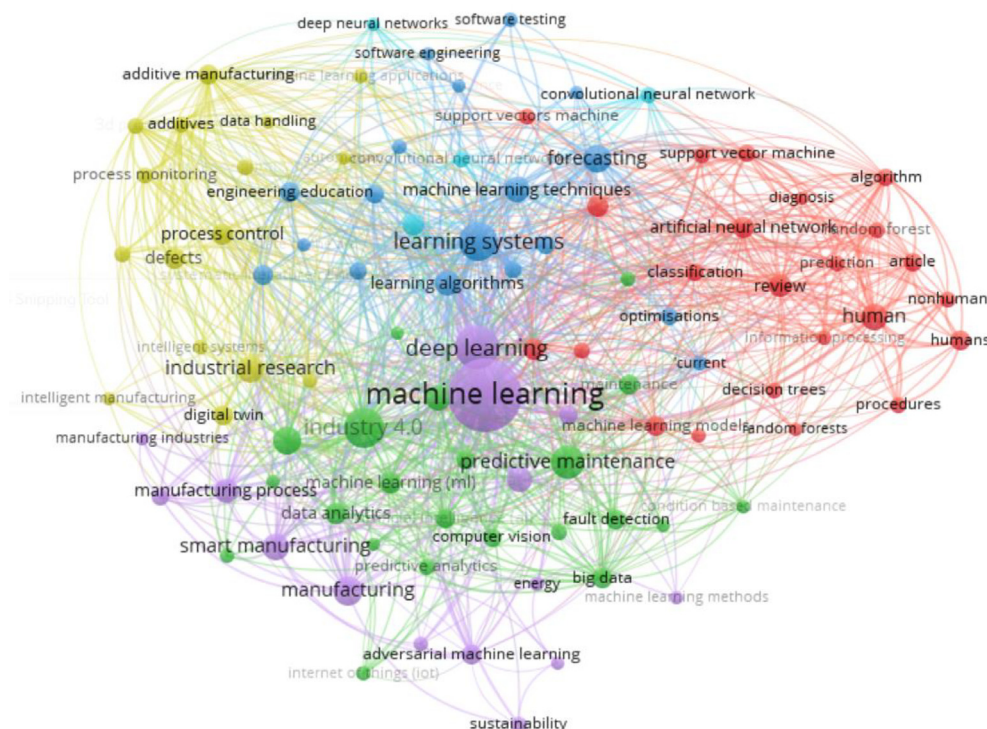
Fig. 1b illustrates the network visualization generated using VOSviewer. Nodes in the visualization represent individual articles or keywords, with their size indicating the relative prominence of the article or theme within the field. Clusters are color-coded based on thematic focus, and edges between nodes represent co-occurrence or citation links. This visualization provides a comprehensive overview of the research landscape, showcasing the relationships between various themes and the distribution of research efforts. For example, the predictive maintenance cluster (Fig. 1b) includes studies leveraging deep learning techniques such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) for machine fault detection and failure prediction. Similarly, the quality control cluster highlights using supervised learning models, such as Support Vector Machines (SVM) and Random Forest (RF), for real-time defect detection. The process optimization cluster focuses on reinforcement learning applications for dynamic production scheduling and resource allocation. The overall data in Fig. 1b reveals several key insights. **First**, predictive maintenance and quality control are dominant research themes, reflecting their critical role in modern manufacturing. **Second**, the connections between clusters highlight the interdisciplinary nature of ML applications, where insights from one domain, such as defect detection, influence advancements in others, like process optimization. **Third**, areas such as real-time adaptive control systems and energy-efficient manufacturing, though less explored, are gaining traction, indicating potential opportunities for future research.

However, the distribution of research efforts, depicted in Fig. 2, reveals that while areas such as fault detection and quality control have been widely explored, other domains like real-time scheduling and adaptive control systems remain relatively underdeveloped. This uneven focus suggests potential opportunities for future research to bridge these gaps and create more integrated ML solutions. Finally, Fig. 3 highlights how research efforts align with specific manufacturing objectives, demonstrating that while predictive maintenance and defect detection are dominant themes, emerging areas such as energy-efficient manufacturing and supply chain optimization are gaining traction. This indicates a shift toward a more holistic application of ML, extending beyond individual machine performance to broader system-level optimizations. Understanding these evolving trends is crucial for identifying future research directions and ensuring that ML applications continue to evolve in alignment with industrial needs.

Numerous investigations including [44–46] were dedicated to data processing in the manufacturing sector. **Hadoop, Spark, Samza, Flink**, and **Storm** are just a few examples of well-known open-source data distribution engines. These engines are adopted in the industry due to their versatility in various contexts and the ease with which their parameters can be adjusted to maximize replica accuracy. In the industrial sector, dealing with massive amounts of data is not uncommon, which requires using ML methods capable of handling high-dimensionality datasets with over 20 attributes. Large-scale ML frameworks, such as **MLlib**, **Mahout**, and **MLbase**, can now be employed.



(a)



(b)

Fig. 1. (a) Annual publication trends (2000–2024) on Machine Learning applications in manufacturing. (b) Network visualization of thematic clusters in ML-driven manufacturing research using VOSviewer.

3. Key components of knowledge implementation in smart manufacturing

Fig. 4 provides an overview of the five essential phases needed to process the data of any smart manufacturing process to create useful information and eventually become knowledge when analyzed and interpreted intelligently. These phases involve data sources, cleaning and manipulation, ML, analysis, and presentation [28]. In the first step, “preparation”, the ML strategy’s goal, available resources, and

limitations are established. The next step in data preparation involves gathering, combining, minimizing, and reformatting the information. Sensors capture data on the initial materials, finished products, and machine settings involved in manufacturing. Merging disparate data sources is what data integration is all about. Cleaning data entails completing blanks, reducing noise, spotting outliers, correcting for data imbalance, and erasing abnormalities. Information loss is kept to a minimum when using data reduction methods like feature selection [7,8,47]. Normalization is one example of a data change that can be

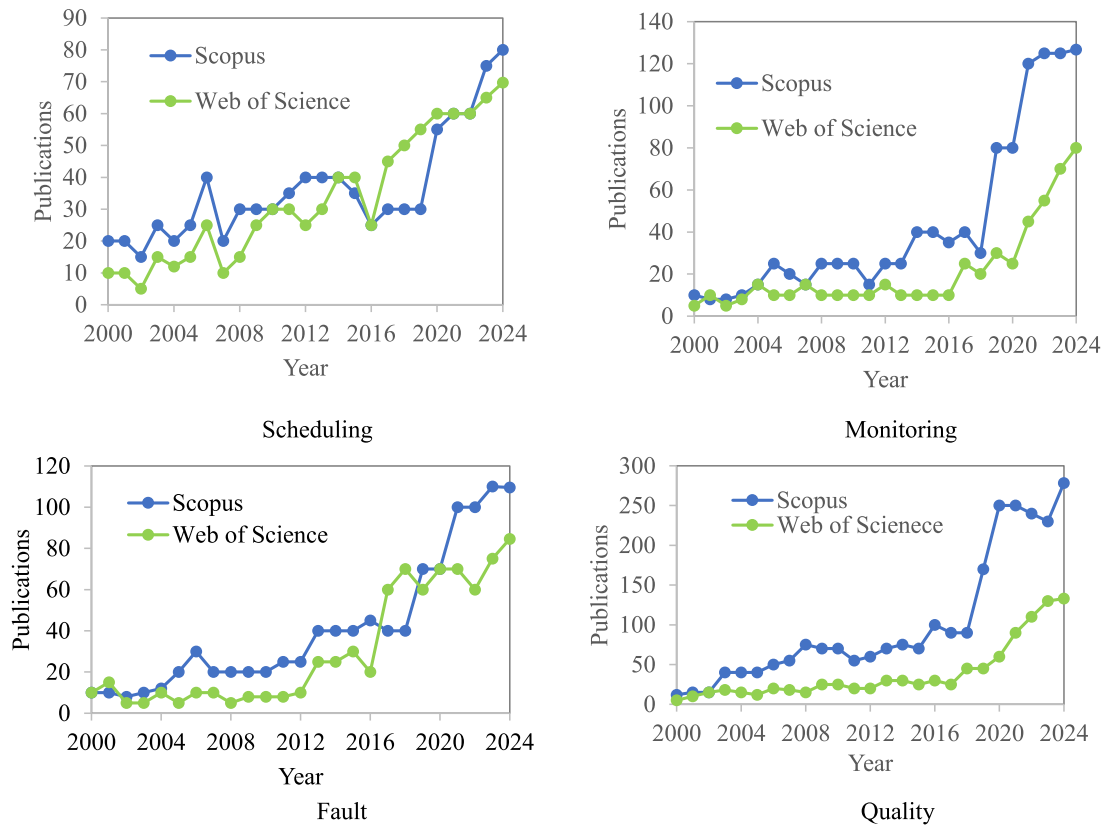


Fig. 2. Scheduling, monitoring, fault detection, and quality control as a percentage of total ML-based publications in Scopus and Web of Science from 2000 to 2024.

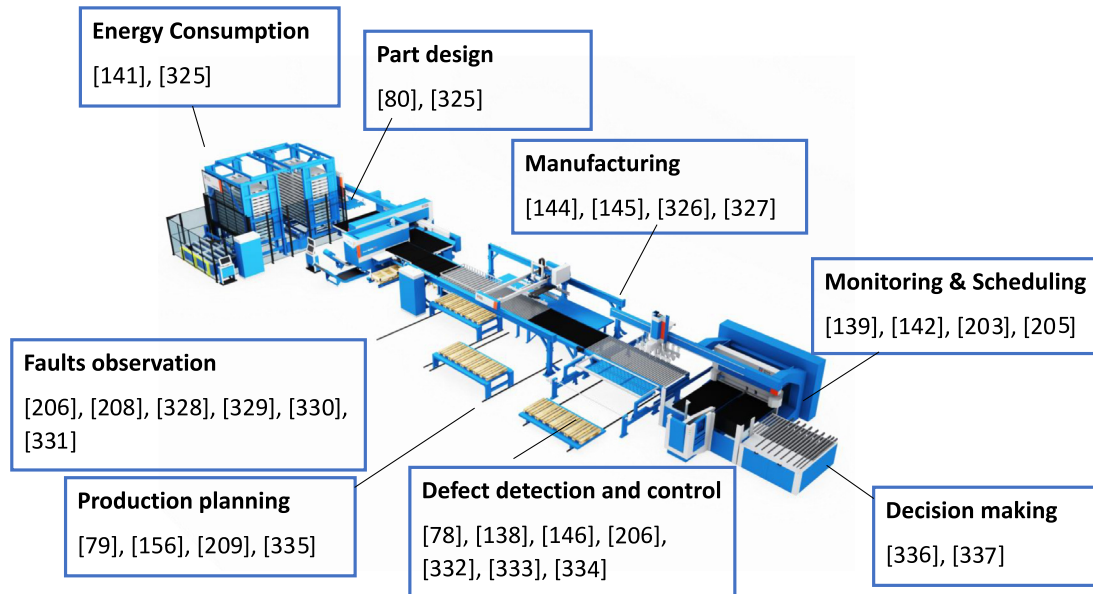


Fig. 3. Manufacturer-focused clustering of ML studies [78–80,138,139,141,142,144–146,156,203,205,206,208,209,325–337].

done before data mining (DM) [48,49]. A data warehouse is where the information resides after it has been cleaned, normalized, and organized.

The third step is to use the proper ML methods on the data in the database to discover regularities, establish rules, or build a model. Un-supervised learning techniques are well-suited for problems that require clustering and association rule mining (ARM) [50,51], as opposed to supervised learning techniques used for classification and regression tasks. The final step involves evaluating the constructed model using

an appropriate performance metric. The final phase analyzes patterns to gauge performance, detect anomalies, or predict outcomes. Refining the knowledge database through multiple iterations may be necessary before achieving satisfactory results. Once the final model is ready, it can be deployed in the manufacturing sector, and regular updates are required to maintain its generalizability with new data. The information database can assist operators and managers in making better decisions or can be incorporated into an automated process to improve the production system itself.

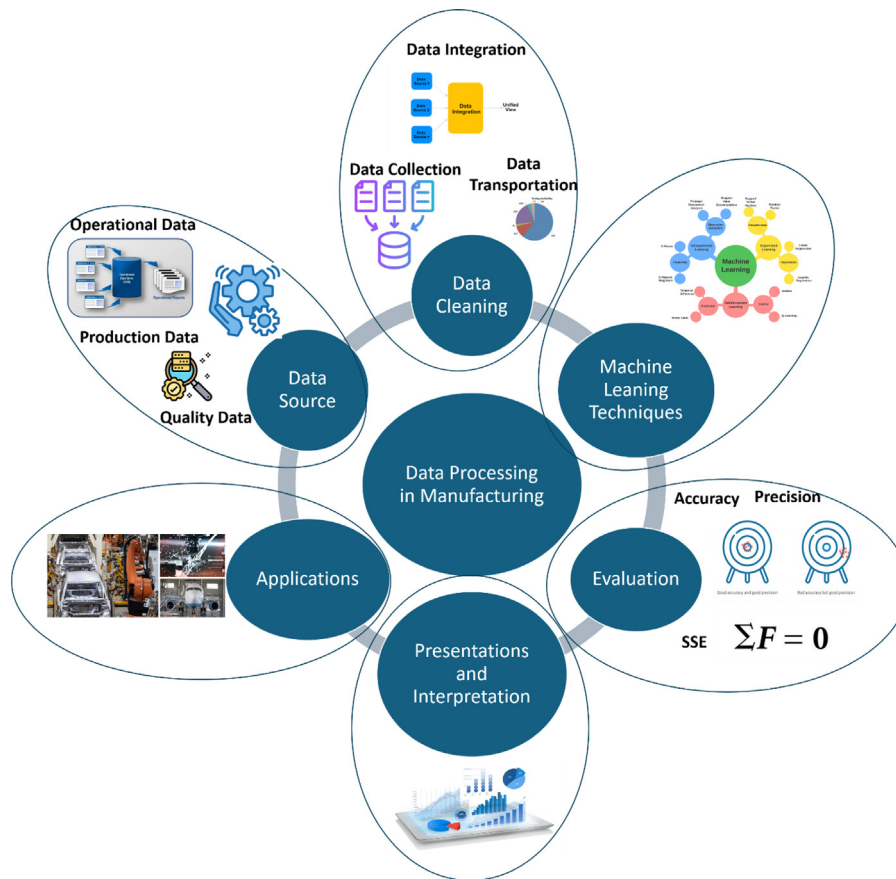


Fig. 4. Typical smart manufacturing knowledge process implementation pillars.

Supervised Learning	Unsupervised Learning	Semi-supervised Learning	Reinforcement Learning
Data scientists provide input, output, and feedback to build the model. Example algorithms: Linear regression - Sales forecasting - Risk assessment Support vector machines - Image classification - Financial performance comparison Decision trees - Predictive analytics - Pricing	Use deep learning to arrive at conclusions and patterns through unlabeled training data Example algorithms: Apriori - Sales functions - Word associations - Searcher K-means clustering - Performance monitoring - Searcher intent Artificial neural networks - Generate new synthetic data - Data mining and pattern recognition	Builds a model through a mix of labeled data, a set of categories, suggestions, and exemplified labels. Example algorithms: Generative adversarial networks - Audio and video manipulation - Data creation Self-trained naïve Bayes classifier - Natural language processing	Self-interpreting but based on a system of rewards and punishments learned through trial and error, seeking maximum reward. Example algorithms: Q-learning - Policy creation - Consumption reduction Model-based value estimation - Linear tasks - Estimating parameters.

Fig. 5. A classification of machine learning methods and their training algorithms.

4. Different ML methods applied in manufacturing studies

Tasks in ML are typically categorized into four types [1,2] (as shown in Fig. 5): supervised, unsupervised, semi-supervised, or reinforcement learning, based on the degree of human supervision involved. While supervised and unsupervised techniques are extensively used in several production settings, progress in semi-supervised and reinforcement learning techniques has been relatively slow. Deep learning has recently gained dramatic attention due to its ability to learn the features directly from the raw data, allowing the performance of complex tasks to improve. Deep learning algorithms or neural networks can be found

in different categories, ranging from supervised learning tasks such as image classification and natural language processing to unsupervised learning through autoencoders. While deep learning is a very efficient process, its serious drawback is its expensive demand for labeled data and computational resources, making some applications very hard to accomplish [6,7,49,50,52]

4.1. Supervised machine learning

Supervised learning is a ML approach where a model is trained using labeled data, meaning the input data is paired with known output

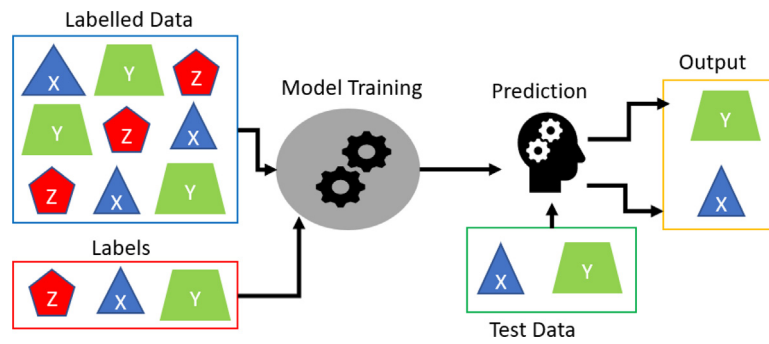


Fig. 6. Schematic of supervised machine learning.

Table 1

An analysis of the literature on supervised machine learning in manufacturing industry (see [68–86]).

Ref	Year	Algorithm						Categories			
		DT	NN	SVM	KNN	NB	Reg.	Scheduling	Monitoring	Quality	Fault
[68]	2024										
[69]	2023										
[69]	2022										
[70]	2022										
[71]	2022										
[72]	2020										
[73]	2020										
[74]	2020										
[75]	2020										
[76]	2019										
[77]	2018										
[78]	2017										
[79]	2017										
[80]	2017										
[81]	2016										
[82]	2014										
[83]	2013										
[84]	2013										
[85]	2012										
[86]	2011										

labels. The model aims to learn the relationship between inputs and outputs, allowing it to predict the correct output for new, unseen data. Fig. 6 shows the process involves feeding labeled input data into the model, training it to make predictions, and then using the model to predict outcomes for new inputs [6,42,53].

The fundamental goal of supervised ML is to use past examples to comprehend the relationship between input and output variables. A model is trained in supervised learning using input variables and their corresponding output variables. The accuracy of predictions a supervised learner makes is directly correlated with the training data it receives. Popular supervised learning applications include classification and regression analysis [31,54]. When predicting categorical data, use classification. When predicting continuous or ordered data, use regression [55]. In economics, classification is a frequently used application [56].

However, conventional mathematical and statistical methods, such as rule-based models and linear regression [31,57], often struggle with handling large-scale, high-dimensional, and non-linear relationships in industrial settings. These traditional methods are typically

limited in adaptability and require extensive manual feature engineering to achieve optimal performance. In contrast, ML techniques, particularly deep learning and ensemble models, offer superior predictive capabilities by automatically extracting relevant features from raw data. Studies have shown that ML models outperform conventional approaches such as fault detection, process optimization, and predictive maintenance by leveraging large, complex datasets with minimal human intervention [29,58]. Additionally, hybrid approaches integrating ML with traditional statistical methods have demonstrated increased accuracy and robustness in real-world industrial applications [59–62].

Table 1 summarizes and compares the research conducted on supervised learning in manufacturing. The table is organized into columns that list the references, the year of completion, the algorithms used on manufacturing datasets, and their corresponding categories. Results from Table 1 reveal that various regression approaches, including NN [10], SVM [63], DT [12], KNN [64] Support vector regression (SVR) [63] and random forests (RF) [65], are frequently utilized in the manufacturing industry. The research found that fault recognition and quality evaluation are the biggest challenges in manufacturing. This is

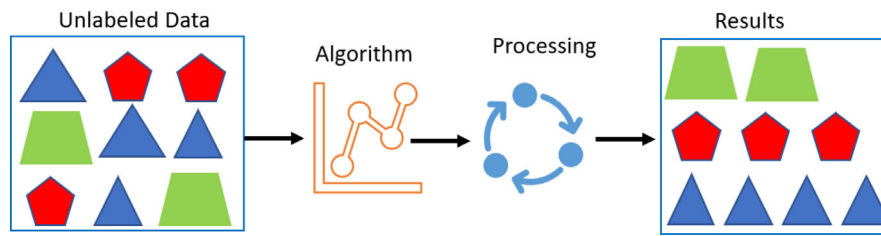


Fig. 7. Schematic of unsupervised machine learning.

also evident in the mainly employed evaluation parameters, in which accuracy and f-measure are typically applied for classification tasks, and R-square statistic is utilized to assess regression models [66,67].

These data preprocessing techniques can be explored for manufacturing data, such as normalization, feature selection, and elimination of missing values [87]. This is the reason behind the complexity due to the heterogeneous and noisy nature of manufacturing data which mainly originates from different sources. It is crucial to ensure the data mining process gets the most out of its information by maintaining good quality data [48]. Therefore, addressing the issues that can arise during the preprocessing stages with produce, processing, and machine variables is essential. The problem of class *imbalance* is significant in the manufacturing industry because it can make the training of ML models less effective. Several researchers have used the SMOTE to tackle this challenge [43]. In the pursuit of improved quality control, many manufacturing facilities record their processing data as time series, which can be used to create forecast models based on massive amounts of sequential data [88].

Supervised ML methods have become commonplace for detecting and addressing errors and failures in the early stages of steel manufacturing [89]. They have also been used in additive manufacturing to develop accurate models [90]. Neural networks are beneficial in this field for developing highly precise predictive models [10]. Many efforts have been directed toward developing ML models that predict quality, as intelligent manufacturing can produce higher-quality products [91]. Certain types of manufacturing raw materials and process equipment can experience wear and tear over their lifetimes, increasing the inherent risks and hazards in production processes. Supervised algorithms have been employed to predict material failures from the inside out within the scope of this discussion.

4.2. Unsupervised machine learning

Unsupervised learning is a ML paradigm aiming to discover unlabeled data patterns and relationships [7], as illustrated in Fig. 7. Unlike supervised learning, the data used in unsupervised learning does not have predefined labels. Unsupervised learning methods include clustering, ARM, and anomaly recognition. The primary goal of these methods is to uncover the underlying structure of the data in a meaningful way without relying on predefined classes. However, research on unsupervised learning has been relatively scarce compared to supervised learning, as manufacturing data often includes class labels, leading to a higher focus on supervised learning [24,92].

Traditional mathematical approaches, such as principal component analysis (PCA) and k-means clustering, have been widely used for pattern recognition and data reduction in industrial applications [93]. However, these methods rely on predefined distance metrics and assumptions about data distribution, which may limit their effectiveness in handling complex, high-dimensional, and dynamic manufacturing environments [28]. Additionally, conventional clustering techniques often struggle with scalability and adaptability when processing large, heterogeneous datasets generated by modern industrial systems.

In contrast, advanced unsupervised learning techniques, including deep clustering, self-organizing maps (SOM), and autoencoders, have demonstrated superior performance in extracting meaningful patterns

from unlabeled industrial data [93–95]. These methods allow for more flexible and adaptive learning, making them highly effective for anomaly detection, predictive maintenance, and process optimization. Studies have shown that hybrid approaches that integrate unsupervised learning with supervised models can further enhance accuracy and robustness in fault detection and adaptive scheduling, providing significant advantages over traditional statistical methods [91,94]. The most important tasks in unsupervised learning are clustering, anomaly detection, and rule mining, which are discussed in detail in the following sections.

Cluster analysis and the detection of abnormalities in manufacturing:

Clustering [95] is a way of grouping similar things. When you look at data, you can group it into clusters depending on how alike the groups are. It requires analyzing the resemblance and dissimilarity of data with the help of specific measurements. Table 2 summarizes the key aspects of manufacturing clustering, including various clustering techniques, their applications, and relevant studies.

There are five primary types of clustering approaches: Different methods are used to group similar data sets, such as partitioning, hierarchical, agglomerative, divisive, and density-based methods [97,102]. Partitioning clustering aims to split the data into k groups, where every group contains highly related data points [103]. Hierarchical clustering [104,105] techniques create cluster trees where the data is grouped according to similarity. In contrast, agglomerative methods merge smaller sets into larger ones to form clusters, and divisive methods break down larger clusters into smaller ones, resulting in groupings that can be analyzed further [106]. The density-based approach scans for areas of greater concentration of data points surrounded by lower concentrations of data points, the clusters are formed with varying shapes and dimensions as a result. Other methods of clustering include grid-based and model-based approaches.

Cluster analysis is a tool that manufacturers commonly use to accomplish tasks such as pattern recognition [107], yield improvement [108], quantitative assessment [109], and equipment condition diagnosis [102]. Clustering has been used in many different fields for different purposes. For example, clustering is used to detect product defects [110], support decisions [111], and layout design challenges [112]. Robots are becoming more helpful in factories as people and machines work together to make things. The rise of robots that can respond independently has prompted the deployment of unsupervised learning models to ascertain the required functions in this field [96].

There are various ways to prepare data for sorting algorithms. A clustering step before the algorithm could improve the performance. The unsupervised approach is commonly used to detect anomalies that may affect the model's accuracy to be trained in a supervised manner [7]. Removing those unusually large or small values will make our predictions more accurate. It makes sense since those extreme values can pull our data in misleading directions. During production, it can sometimes be essential to discover the defects in the given data. For that, clustering can help us out because it helps identify those defects before traditional classification algorithms. It is said that clustering algorithms help analyze many production data sets to evaluate how much similarity can be found among them. Therefore, its decisions are also recognized as necessary for businesses, which help them create various

Table 2
Clustering techniques used in manufacturing.

Aspect	Details	Key Studies
Clustering	1. Partitioning: Divides data into k groups. 2. Hierarchical: Creates a tree structure of clusters. 3. Agglomerative: Merges smaller clusters. 4. Divisive: Splits larger clusters. 5. Density-Based: Identifies clusters based on the density of data points	[95] [96,97]
Techniques	- K-means: Partitions data into k clusters based on feature similarity. - Hierarchical Clustering: Builds a hierarchy of clusters. - Density-Based Spatial Clustering of Applications with Noise (DBSCAN): Density-based clustering that identifies clusters based on high-density areas. - Self-Organizing Maps: Neural network-based approach to clustering.	[98,99]
Applications	- Pattern Recognition: Identifying trends and patterns in manufacturing data. - Yield Improvement: Enhancing production output by analyzing cluster data. - Quality Control: Detecting defects by clustering similar products. - Layout Design: Optimizing factory layouts based on product flow patterns.	[100,101]

Table 3
Anomaly detection techniques used in manufacturing.

Aspect	Details	Key Studies
Types	1. Density-Based: Detects anomalies based on the density of data points. 2. Clustering-Based: Identifies outliers within clustered data points. 3. Model-Based: Uses statistical or ML models to identify deviations.	[115,120]
Techniques	- K-Nearest Neighbors (KNN): Identifies outliers by analyzing the distance to nearest neighbors. - Local Outlier Factor (LOF): Evaluates local density to detect anomalies. - Support Vector Machines (SVM): Uses hyperplanes to separate normal data from outliers.	[121,122]
Applications	- Process Monitoring: Detects anomalies in manufacturing processes to ensure quality. - Quality Control: Identifies unexpected deviations in product quality. - Fault Detection: Alerts operators about abnormal conditions in machinery or processes	[24,115,123]

marketing campaigns, policies, and goals. Cluster quality is evaluated by using diverse evaluative parameters like accuracy and completeness. These clustering algorithms use Different criteria to approve cluster output. The goal is to develop clustering moths that work optimally on real data and to implement algorithms that do not compromise the overall performance. Usually, these algorithms are judged by their in executing tasks using their measures and standards and comparing their performance to other algorithms [113,114].

Anomalies, also known as outliers, are objects that may not be immediately visible as they do not conform to the established pattern of the dataset to which they belong [115,116]. Detecting such occurrences is a crucial task in ML, and it involves three primary categories of techniques: those that rely on density, clustering, and SVM. Examples of these techniques include the KNN algorithm and the local outlier factor [117]. Table 3 summarizes the key aspects of manufacturing anomalies, including various clustering techniques, their applications, and relevant studies.

Data mining is valuable for discovering manufacturing system patterns and making monitoring and inspecting processes easier. It can help detect outliers in process values or unexpected quality outcomes. If the algorithm identifies an issue, it can notify the appropriate recipient, such as an operator or system, to take the necessary steps to resolve the problem. Several performance measures are used to evaluate the

quality of the results obtained from anomaly detection algorithms [48,49,118,119].

Table 4 comprises the references, publication years, study objectives, and employed methods in recent and significant publications concerning clustering and anomaly detection. Furthermore, Table 5 provides examples of studies that used data from the semiconductor manufacturing industry, as presented in [115,124–126]. Unsupervised learning strategies are frequently employed in the manufacturing industry to improve manufacturing processes by materializing monitoring tasks. Today, in the manufacturing industry, clustering methods such as K-means, hierarchical clustering, DBSCAN, and self-organizing maps are widely used to identify hidden patterns and structures within the data [95,105,106]. The success of clustering is measured using objective metrics such as the percentage of expenses avoided, the number of derived rules, and minimal defect coverage. Unsupervised learning methods are also used in manufacturing and have become increasingly popular in recent years. Many studies have shown that unsupervised learning methods can discover hidden patterns in data just as effectively as supervised learning methods [7,92]. In complex data scenarios, feature selection methods like principal component analysis (PCA) have proven to be the most efficient approach to achieving desired outcomes while also speeding up the algorithm [93,127]. Principal component analysis helps reduce the number of features

Table 4

Manufacturing case studies involving cluster analysis and anomaly detection (see [128–147]).

Ref	Year	Algorithm						Categories			
		K-means	DBSCAN	Hierarchical	SOM	LOF	Others	Scheduling	Monitoring	Quality	Fault
[128]	2024										
[129]	2023						PCA				
[130]	2023						PCA				
[131]	2022										
[132]	2021										
[133]	2020						RF				
[134]	2019										
[135]	2019										
[136]	2019										
[137]	2018										
[138]	2018										
[139]	2018										
[140]	2018										
[141]	2017										
[142]	2017										
[143]	2017										
[144]	2017						PCA				
[145]	2017										
[146]	2016										
[147]	2014										

Table 5

Examples of manufacturing case studies using association rule mining (see [148–165]).

Ref	Year	Algorithm				Categories			
		Apriori	Fp-Growth	Eclat	Others	Scheduling	Monitoring	Quality	Fault
[148]	2024								
[149]	2023								
[150]	2022								
[151]	2021								
[152]	2020								
[153]	2020								
[154]	2019								
[155]	2019								
[156]	2018								
[157]	2018								
[158]	2017								
[159]	2017								
[142]	2017								
[160]	2016								
[161]	2015				PCA-WARM				
[162]	2015				SLP-Growth				
[163]	2015								
[164]	2013								
[165]	2013								

used in clustering by identifying the most significant variables that contribute to the underlying patterns in the data. Clustering techniques and feature selection methods reduce computational burden, improve accuracy, and provide insights for manufacturers to increase efficiency and reduce costs [95,110].

Association- and sequential-rule mining in manufacturing:

One of the primary objectives of data mining is to uncover patterns of association among the objects in a large dataset, often achieved

through Association Rule Mining (ARM). ARM algorithms include Apriori [166], FP-Growth [167], and Eclat [168].

The sequential pattern mining (SPM) technique aims to discover recurring sequences of item sets in a dataset and uncover cyclic correlations between variables. It searches for informative subsequences that are hidden within larger sets of sequences. To evaluate the level of interest in a sequence, SPM compares it to a set of relevant criteria, such as occurrence frequency, length, and profit. Commonly used SPM methods include PrefixSpan, Generalized Sequential Pattern (GSP),

Table 6
Types, techniques, and applications of ARM and SPM in manufacturing.

Aspect	Details	Key Studies
Types	1. ARM 2. SPM	[50,171]
Techniques	ARM Techniques: - Apriori: Mines frequent itemsets and generates association rules. - FP-Growth: Efficiently mines frequent patterns using a tree structure. - Eclat: Uses vertical data format for mining. SPM Techniques: - PrefixSpan: Efficient mining of sequential patterns. - Generalized Sequential Pattern (GSP): Finds generalized patterns in sequences. - SPADE: Density-normalized event mining using vertical format.	[166,170]
Applications	-Production Scheduling: Optimizing manufacturing workflows. -Quality Improvement: Enhancing product quality through insights. -Machine Performance Analysis: Identifying performance trends. -Safety Hazard Detection: Identifying factors leading to potential hazards. -Environmental Sustainability: Evaluating and reducing ecological impact.	[172,173]

and Spanning-Tree Based Clustering for Density-Normalized events (SPADE) [169]. Table 3 presents notable recent works in SPM and industrial ARM. Statistical research shows that Apriori is the most commonly used algorithm. This technique has been applied in various areas of the manufacturing industry, including production scheduling, text mining, analysis of machine performance, quality improvement and others [170]. Table 6 summarizes different types, techniques and applications of ARM.

ARM methods are commonly used in the manufacturing industry as they have proven efficient and effective. These methods have not only improved production speed but also helped create high-quality products while minimizing environmental damage. As a result, manufacturing has become more cost-effective, and the workplace has become safer. To prevent accidents at work, some companies use arm models to identify complex links between different factors [174]. These models analyze data from various sensors to determine factors such as temperature and humidity, as well as the intensity of light. This helps identify potential hazards that could lead to accidents in the workplace. By closely monitoring these factors, businesses can prevent accidents from occurring, which is highly effective in reducing the number and severity of accidents in many workplaces. In addition, these predictive safety models can provide manufacturers with helpful information on detecting potential safety hazard concerns in the work environment. This will subsequently allow them to build a safer and healthier workplace. Reducing pollution levels is another important focus point for companies to offset some environmentally negative aspects of industrial practices. Can manufacturers identify the relationship between automation and air quality to develop eco-friendly practices, this is an important question that must be asked. Manufacturers can utilize the capacities of ARM to evaluate and lower their ecological footprint [175]. Somehow, ARM technology has helped discover other challenging mechanical works, for instance, the creation of solar power. Identifying crucial connectivity or structure between different elements results in an improved product standard. Using ARM technology for identifying anomalous event patterns can help us keep track of production processes. Anomalous patterns may suggest potential faults or deviations from the standard workflow if not closely monitored and acted upon. This use of ARM can prove vital in ensuring quality control and safety standards are maintained in such areas. By identifying and addressing any outliers and abnormalities, ARM delivers an early warning system for manufacturers to catch and resolve problems before they delay production [50].

In summary, it has been evidenced that the ARM methods can play a crucial role in the manufacturing industry by improving safety, promoting environmental sustainability, and enhancing the quality of the products. ARM Models benefit manufacturers by avoiding safety hazards and developing eco-friendly practices while producing more reliable and long-lasting goods. By detecting abnormalities and outliers, ARM aids in enhancing the monitoring process to make production processes more efficient and improve productivity.

4.3. Deep learning

Deep Learning (DL) refers to developing neural networks that consist of multiple hidden neurons, as shown in Fig. 8. This widely recognized definition reflects the successful application of DL techniques in various academic fields and the business world [176]. Gradient descent is commonly used to train these models and learn appropriate NN weights. Although the architecture of deep neural networks requires the selection of several hyperparameters, this ML technique holds great promise for many manufacturing applications. Deep learning techniques have found successful applications in various manufacturing processes, such as improving condition prognosis [177], predicting and enhancing quality [23], analyzing the root cause of failures [178], and optimizing inspection procedures [179]. The development of hybrid techniques combining fuzzy logic and neural networks has been driven by the unpredictability inherent in manufacturing [180,181].

The industrial sector favors Convolutional Neural Networks (CNNs) for deep learning [182,183]. Deep Recurrent Neural Networks (DRNNs) aid in the prediction of droplet evolution and understanding the process dynamics [184]. Another neural network architecture is the Deep Boltzmann Machine, DBM, with a variant called the Restricted Boltzmann Machine, RBM, which can benefit makers if implemented appropriately [185]. A deep belief network (DBN) is a complex system comprising many neural grids with hidden levels. Understanding the DBN requires knowledge of the belief net and the rational choice model (RBM). It is the process of examining characteristics and their relationships for different ranges [186]. DBN has been applied to several operations in the industry by researchers, such as quality prediction [77], prediction in induction motor failure [78], and so on. A deep autoencoder is formed when the asymmetrical DBNs, one to encode and another to decode, each have four or five layers. DA code has

Table 7

Select case studies of deep learning's use in industry (see [176,189–209]).

Ref	Year	Algorithm					Categories			
		CNN	DNN	DBM	DEN	DA	Scheduling	Monitoring	Quality	Fault
[189]	2024									
[190]	2023									
[191]	2023									
[192]	2023									
[193]	2022									
[194]	2022									
[195]	2022									
[196]	2021									
[197]	2021									
[198]	2020									
[199]	2020									
[176]	2020									
[200]	2019									
[201]	2019									
[202]	2019									
[203]	2019									
[204]	2018									
[205]	2018									
[142]	2017									
[206]	2017									
[207]	2017									
[208]	2017									
[209]	2017									

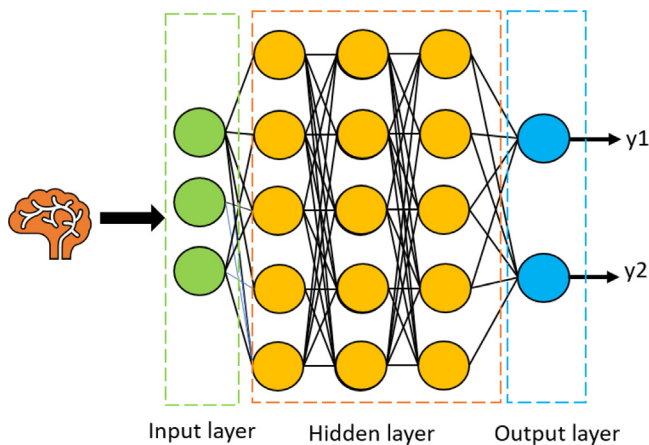


Fig. 8. Deep machine learning principle.

been applied to manufacturing jobs using bearing health analysis, data-driven fault diagnosis, and wafer fault detection [187,188]. Table 7 outlines some recent deep-learning research in manufacturing.

Deep learning methods offer several advantages over traditional models, including fewer limitations and no need for generalizations in everyday operations [210]. Additionally, deep learning can help lower production costs and enable automatic defect classification, improving overall efficiency and productivity [211]. Using deep neural networks (DNN) in transporting fragile products has enabled safer transportation [212]. In the digitalization of production processes, CNN has been utilized to forecast the time-of-flight of radio burst indicators, enabling more accurate and efficient operations. There is a significant need

to devise efficient techniques for monitoring the electromechanical systems applied in intelligent manufacturing. It has been found that deep-learning-based autoencoders provide a good deal of support during the operation of such pivotal models [213]. Deep learning has thus far been proven powerful within the manufacturing industry. It has enabled researchers to meet various objectives that improve the quality of various manufacturing processes. As summarized in Table 8, DL methods enhance efficiency, reduce costs, and improve safety in manufacturing environments. Additionally, hybrid approaches combining fuzzy logic and neural networks have shown promise in handling the unpredictability inherent in many manufacturing tasks.

4.4. Semi-supervised learning

Semi-supervised learning (SSL), as shown in Fig. 9, is a powerful approach that integrates small amounts of labeled data with large quantities of unlabeled data to improve model performance [102]. Semi-supervised learning is particularly valuable in manufacturing because labeling data in complex industrial environments requires substantial costs and effort. Semi-supervised learning techniques can thus improve model accuracy and robustness while requiring fewer labeled datasets [221].

One notable application of SSL in manufacturing is predictive maintenance, where semi-supervised models can use machine sensor data to predict failures with minimal labeled failure data, reducing the reliance on costly manual inspections [102]. SSL is also employed in quality control, where it can help identify product defects by learning from a small set of labeled defective items and a large amount of normal product data, thus improving defect detection without the need for extensive manual labeling. Furthermore, SSL has been effectively applied in optimizing supply chain processes, where limited historical

Table 8
Summary of deep learning applications in manufacturing.

Method	Advantages	Disadvantages	Key Applications
CNNs	- Excellent at handling image-based data (e.g., defect detection) - Highly accurate for visual tasks - Automatic feature extraction	- Requires large labeled datasets - Computationally expensive, especially with deeper architectures	- Defect Detection: Visual inspection in production lines [182] - Quality Control: Automated image-based product analysis [214]
DRNNs	- Ideal for time-series data - Can handle sequential data such as process dynamics and event prediction	- Prone to vanishing gradient problem with long sequences - Challenging to train compared to feedforward networks	- Process Monitoring: Predicting droplet evolution in manufacturing [184] - Predictive Maintenance: Anticipating machine failures [215]
DBMs & RBMs	- Good for learning complex, hidden patterns - Can be applied to both supervised and unsupervised learning tasks	- Difficult to train due to slow convergence - Computationally intensive	- Fault Detection: Identifying defects and failures in manufacturing systems [216] - Anomaly Detection: Detecting irregularities in processes [217]
DBNs	- Can capture hierarchical data relationships - Suitable for feature learning and classification tasks	- Requires large computational resources for training - Can be sensitive to noisy data	- Quality Prediction: Product quality control in manufacturing [218] - Motor Failure Prediction: Fault diagnosis in motors [187]
Autoencoders	- Effective for dimensionality reduction and unsupervised learning - Good for detecting anomalies and faults	- Can overfit with small datasets - Requires careful tuning of hyperparameters	- Fault Diagnosis: Identifying anomalies in bearings and other critical components [219] - Health Monitoring: Monitoring machine health in real time [220]
Hybrid Techniques (Fuzzy Logic + Neural Networks)	- Combines strengths of both systems (handling uncertainty and deep learning's prediction power) - Useful in complex, unpredictable environments	- Can be difficult to design and tune - Performance may degrade with high levels of noise	- Electromechanical System Monitoring: Optimizing intelligent systems in manufacturing [176] - Process Optimization: Improving system efficiency under uncertainty [180,181]

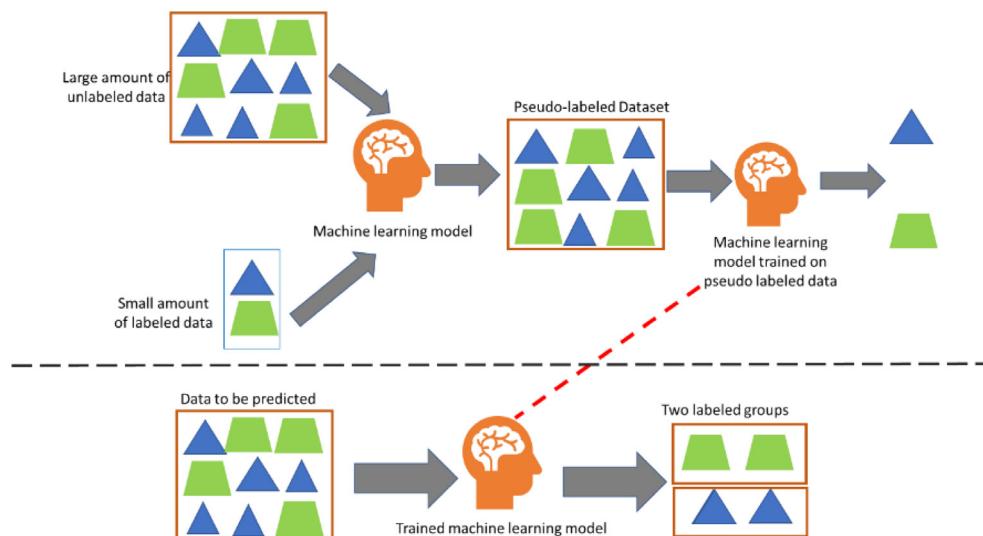


Fig. 9. Schematic of semi-supervised machine learning.

data can predict demand fluctuations and optimize inventory levels [108].

However, there are challenges in applying semi-supervised learning in manufacturing. One significant issue is the risk of overfitting, where the model may excessively rely on the limited labeled data and fail

to generalize well to unseen data. Furthermore, suppose the unlabeled data does not sufficiently represent the underlying process or system. In that case, SSL models may perform poorly, learning patterns that do not accurately reflect real-world conditions. Despite these challenges, SSL remains a promising technique, especially in resource-constrained

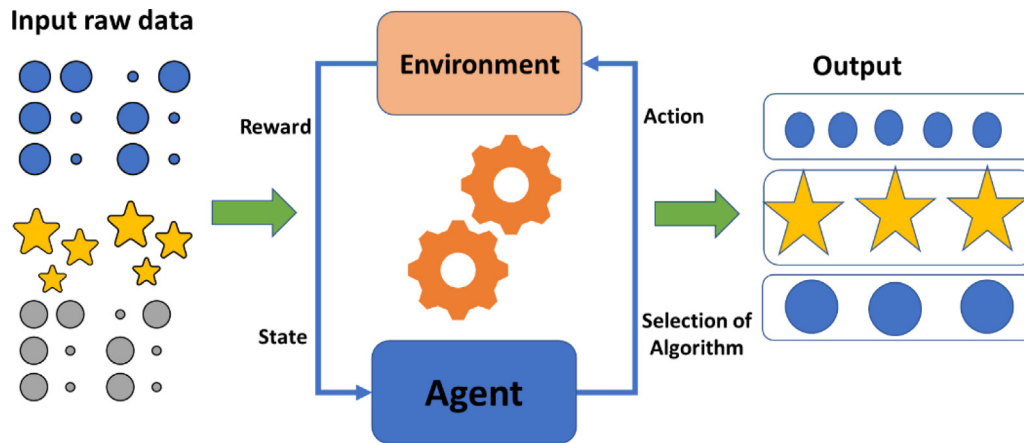


Fig. 10. Reinforcement machine learning concept.

environments, as it enables manufacturers to derive meaningful insights from limited labeled datasets, reducing costs and improving operational efficiency [221,222].

One example of an easy-to-understand approach for SSL is self-training. First, the available labeled data are used to train a model. This model is then employed to assign labels to the unlabeled ones, which become pseudo-labels in this case. The dataset of labeled and pseudo-labeled examples is used to retrain the model. This process of iterations continues until the model's performance stabilizes. For instance, in manufacturing applications, self-training could be applied in fault detection tasks where there are few instances with known faults but many unknown cases. Co-training is a process of training several models on different parts of the data or its representations. Each model predicts labels for unlabeled objects, and these predictions are used to teach other models. To enhance overall performance, the idea is to utilize distinct but complementary information from different learners. Co-training becomes handy in manufacturing when it involves multiple feature sets or sensor data, enabling learning from various perspectives and improving prediction accuracy [222,223].

With graph-based methods, each data point is treated as an entity in a graph, whereby edges signify relationships between nodes. Some techniques include label propagation and graph-based SSL, which apply this structure to propagate labels among labeled and unlabeled nodes, respectively. Consequently, similar data points can be assumed to share common labels, while graphs' relationships can be exploited for better learning. Process control optimization and predictive maintenance enhancement in manufacturing may be done using graph-based methods that capture intricate patterns within the dataset [224].

Consistency regularization encourages models to make predictions that are the same for augmented versions of the input. This helps the prediction remain the same when input values change, making it generalize better. In pseudo-labeling, labels are assigned to unlabeled data using the model's prediction. Subsequently, these pseudo-labels are used in training to fine-tune the model. This way, more labeled-like examples are obtained through pseudo-labeling, leading to accurate and robust models. Pseudo-labeling becomes an innovative solution when obtaining labeled data is difficult, such as in complicated manufacturing processes. Multi-view learning exploits multiple representations or feature sets of the data to improve learning. Each view has supplementary information, and models are trained on these different views to enhance overall accuracy. This method is useful in manufacturing when various types of sensors or data modalities are present, allowing for integrating different information sources for improved accuracy and robustness [225].

Transductive Support Vector Machines (TSVMs) incorporate unlabeled data into the training process, which takes traditional Support Vector Machines (SVMs) a step further. Unlike inductive SVMs, which

generalize over unseen examples, TSVMs concentrate on maximizing performance for specific unlabeled examples seen during training. Using unlabeled data helps refine decision boundaries. In manufacturing, TSVMs can be utilized for fault detection and process optimization applications with plentiful unlabeled data [226,227]. Semi-supervised learning provides a useful way to deal with the problems of limited labeled data in manufacturing. By using different SSL methods, such as self-training, co-training, graph-based approaches, multi-view learning, and TSVMs, manufacturers can use unlabeled data to boost model performance and make operations more effective. As manufacturing systems become more complex, SSL techniques will play a key part in improving data-driven choices and improving production processes overall [228].

4.5. Reinforcement learning

Reinforcement learning (RL) has a revolutionary effect on manufacturing. It allows manufacturers to improve complex processes through ongoing learning and adjustment [229–231]. The main idea of RL involves an agent learning to act in an environment to get the most rewards over time (Fig. 10). This means trying different strategies and changing based on what happens. Reinforcement learning differs from supervised learning, which needs old labeled data. Instead, RL makes choices by interacting with its surroundings and learning from rewards and punishments. RL is a good fit for changing manufacturing situations where old methods might not work well [230–232]. For instance, RL has been successfully applied in adaptive production scheduling, where machines and robots can optimize their actions in real-time based on current conditions and production requirements, improving efficiency and reducing downtime. RL has also been used for energy optimization in manufacturing plants, where it can dynamically adjust processes to minimize energy consumption while maintaining product quality and production targets. Additionally, RL has been applied to inventory management, helping businesses manage stock levels by learning optimal restocking strategies from historical data and operational conditions, thus reducing waste and improving resource utilization [233,234].

However, applying RL in manufacturing is not without its challenges. One issue is the requirement for large amounts of data to train RL models effectively, which can be challenging to obtain in industrial settings. Additionally, the reward function in RL may need to be carefully designed to align with business objectives, and an improper reward function could lead to suboptimal performance. Furthermore, RL models can be computationally expensive and time-consuming to train, which might limit their applicability in some resource-constrained environments. Despite these challenges, RL's ability to continuously adapt and optimize processes makes it a promising tool for modern manufacturing systems [233,235].

Table 9
Reinforcement learning (RL) techniques in manufacturing.

Method	Advantages	Applications
Q-Learning	- Easy to implement - Efficient for small state-action spaces	- Production Planning: Optimizing machine settings for cost reduction and output maximization [236]
DQN	- Handles large, complex state spaces - Suitable for high-dimensional problems	- Robotic Automation: Automating production lines in complex environments [239]
PG	- Works well in continuous action spaces - Fine-tunes actions for better control	- Adaptive Control: Fine-tuning manufacturing controls in real-time [240]
AC	- Improves stability and learning speed - Suitable for complex optimization problems	- Supply Chain Management: Optimizing supply chain processes and energy management [241]
MARL	- Facilitates experience sharing between agents - Enhances overall system performance	- Distributed Manufacturing Networks: Coordinating systems in distributed production environments [232]

Q-learning is a key method in RL. It helps agents learn a value function called the Q-function. This function shows how useful taking a certain action in a given state is. The agent keeps updating this function based on the rewards, which helps it find the best way to get the most rewards. In factories, Q-learning can help improve production plans or tweak machine settings to boost output and cut costs. Table 9 overviews various RL techniques and their applications in manufacturing environments. Deep Q-Networks (DQN) is another well-known approach. It blends Q-learning with deep neural networks to deal with complex state spaces [236]. DQN lets RL agents learn tricky policies using a neural network to estimate the Q-function. This makes it a good fit for tasks with big, complex state spaces, like advanced robots and automated production lines [237,238].

Another class of RL techniques that operate on directly optimizing the policy used by an agent is the Policy Gradient method. They also acquire the capability to solve the problems related to policy parameters concerning the gradient of expected reward and apply it to many non-discrete action spaces. Since policy gradient emission is related to actions, it is inherited in manufacturing applications where actions are to be fine-tuned, for example, in adaptive control or real-time control. These methods contain both value-based and policy-based methods and have some advantages. Eventually, these methods employ two different models: an actor responsible for choosing policies and a critic who rates measures implemented by the same actor [242]. The actor leverages the critic's correction to enhance its policy, which improves stability and fast learning. Actor-critic methods can be employed in manufacturing applications with complex processes to be optimized, such as energy or supply chain management. Multi-agent reinforcement learning builds on traditional reinforcement learning to address scenarios with many interacting agents. Each agent learns its policy while also adopting the information present in other agents. This applies in collaborative and competitive settings, such as coordinated manufacturing systems and distributed production networks. Multi-agent reinforcement learning can allow agents to achieve higher performance more quickly than an individual agent could on its own by facilitating the sharing of experiences that can help improve overall system performance and overall agent adaption of helpful experiences [243].

Reinforcement learning is a useful paradigm to consider when faced with our complex and dynamically evolving manufacturing problems. With techniques like Q-learning, Deep Q-networks, Policy Gradient approaches, Actor-Critic approaches, and Multi-agent reinforcement learning, firms enhance the efficiency in process improvement, quality

management, and predictive maintenance. This means that as manufacturing environments increase in their level of sophistication, reinforcement learning will be one of the elemental components that can spur innovation and improve performance [230,233,234].

5. Implications regarding the selection of appropriate ML algorithm

Each “knowledge discovery” episode is marked by a distinct set of circumstances, rendering the search for the optimal ML approach to address manufacturing challenges an ongoing pursuit. Expert professionals with relevant experience are better equipped to propose a strategy based on previous studies. However, it is not easy to guarantee the success of their recommendations in a specific manufacturing environment. To effectively address a particular manufacturing issue, the success of any algorithm is heavily dependent on the quality of available data and the decisions made when configuring the algorithm's parameters. Researchers must be knowledgeable of the algorithms published in relevant academic literature and should evaluate their effectiveness in real-world manufacturing applications. Conversely, producers primarily working in the production industry may require a broader understanding of ML methods and applications, creating an additional obstacle during manufacturing. ML experts may also need more familiarity with the manufacturing field to successfully apply their skills, which can present challenges to completing their work. Moreover, to achieve the best possible outcomes for a given manufacturing issue, exploring a range of approaches for identifying optimal parameter settings for each algorithm is recommended. This challenge is significant and spans all manufacturing studies. Since several iterations of an algorithm may need to be run or its parameters investigated, integrating ML strategies into regular factory operations may be time-consuming. Therefore, a combination of various ML approaches should be considered for a single investigation. Additionally, to further the model's flexibility, it should be constantly updated with the most recent data [28,61,244–287].

A direct comparison among the various ML models and frameworks used in manufacturing can shed light on their strengths, limitations, and suitability under different tasks. This section compares the overall suitability of model options reviewed in Section 4, side by side in select application categories, to better assist practitioners in reaching a decision on the optimum selection that may suit their particular ML problem.

Supervised versus Unsupervised Learning in Quality Control:

Quality control in manufacturing can be based on various supervised and unsupervised learning techniques. Among all other techniques, Support Vector Machines and Random Forests are widely [65, 227, 261] used when pre-labeled data is available for the correct classification of defects. However, these models may require massive labeled datasets to provide high performance, which is highly costly and time-consuming to acquire. On the other hand, unsupervised models, in this case, clustering algorithms like K-means, enable anomaly detection without using labeled data and are effective during the early stages of quality control. It has also been noted that, in general, even though supervised models have given assurance over higher accuracy in artificial environments, unsupervised methods grant greater flexibility and adaptability in dynamic real production environments where labeled data is usually scarce [128].

Deep Learning Compared to Traditional ML Models in Image-Based Defect Inspection:

Deep learning models such as CNN are useful in image-based inspections within industries like electronics and automotive [214], where the reason is that traditional models lag behind the power of feature extraction from complex image data by CNNs. For instance, CNNs could detect microscopic defects in circuit boards with more than 90% accuracy [262]. At the same time, traditional methods rely greatly on manual feature engineering and generally report lower results in terms of accuracy. However, using CNNs is very computationally expensive, requiring large datasets and not being feasible in every manufacturing setting. In the case of limited data availability or computational resources, traditional models may be the more viable alternative; for example, decision trees and SVMs are generally low in processing demands yet provide acceptable accuracy [12, 13].

Reinforcement Learning versus Rule-Based Systems for Process Optimization:

The optimization of manufacturing processes might be regarded as reinforcement learning or rule-based systems. The former is well adapted for complex dynamic environments where a strategy or policy is learned through experiences and actions [229, 230, 255]. For example, reinforcement learning applied to chemical manufacturing optimizes reaction parameters in near real-time, continuously adapting to assure product quality [263]. In contrast, rule-based systems are often not adaptive [57], whereas more complex and expensive-to-implement RL has to be updated manually in case of changes in process conditions. It has been established through comparative studies that rule-based systems may work efficiently in static environments where process conditions can be stipulated. However, RL provides a dynamic, scalable solution when process conditions change quite frequently [229].

Accuracy, Scalability, and Interpretability as Comparison Criteria:

Generally speaking, ML model selection within manufacturing applications represents a trade-off among these three aspects: model performance, scalability, and interpretability.

Accuracy: Deep learning models, including CNNs and RNNs, assure very high accuracy in many applications that require high-dimensional data processing, such as in visual inspections or predictive maintenance [182, 183, 264].

Scalability: The unsupervised and reinforcement learning models may become more scalable since they require less labeled data and can adapt to various applications. However, they may be less precise than the supervised ones in some well-defined tasks [7, 8, 229].

Interpretability: Traditional ML models, such as Decision Trees, are usually more interpretable, which allows manufacturers to understand model decisions and instill trust in them. Sometimes, this comes with the cost of reduced accuracy or flexibility because deep learning models often represent “black-box” approaches [12, 176].

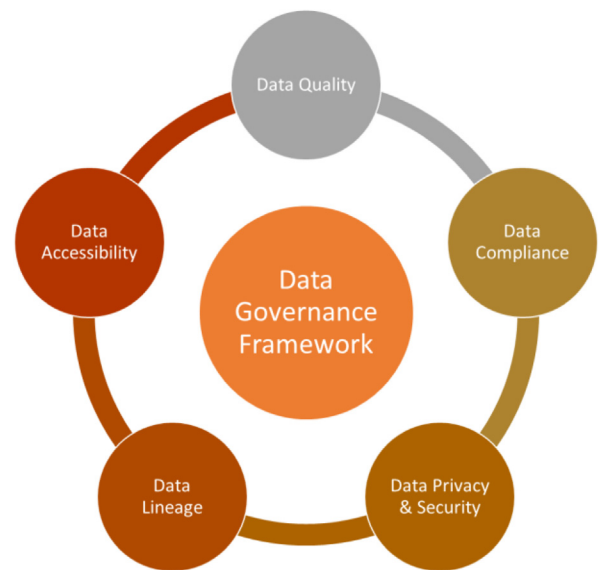


Fig. 11. Data governance framework and pillars.

6. Ethical considerations and data privacy in manufacturing ML applications

With the increasing integration of ML technologies into the manufacturing systems, it is equally important to have ethical considerations and data governance frameworks to guarantee responsible utilization. While ML promises enhanced productivity and efficiency, it also brings challenges in data privacy, security, bias, and accountability. In this section, these key concerns are discussed, and potential strategies for addressing them are outlined — drawn from practical examples reported in the literature.

6.1. Data privacy and governance in manufacturing

Manufacturing processes generate vast amounts of data, often encompassing proprietary information, employee data, and sensitive operational details. Effective data governance is crucial to protect this information and ensure compliance with regulations such as the General Data Protection Regulation (GDPR) [5] in Europe and the California Consumer Privacy Act (CCPA) [288] in the United States. These laws mandate responsible data collection, processing, and storage practices, providing individuals and organizations with specific rights over their data.

Data governance in manufacturing must incorporate strong encryption methods and strict access controls to safeguard sensitive information. Encryption is essential both in transit and at rest, while access controls ensure that only authorized personnel can access specific datasets. A well-structured Data Governance Framework should include key components such as data quality management, privacy measures, and access controls. Developing robust incident response plans for potential data breaches is critical [5, 288]. These plans should outline procedures for notifying affected parties, mitigating damage, and ensuring regulatory compliance. Fig. 11 illustrates an overview of data governance framework.

6.2. Algorithmic bias in ML models

Another major ethical challenge in ML deploying across different industries, including manufacturing, is algorithmic bias. The different sources of bias in ML models can justify data historical bias, sampling bias, or label bias, in which there is subjective human judgment influencing data labeling [248]. In manufacturing, bias may influence

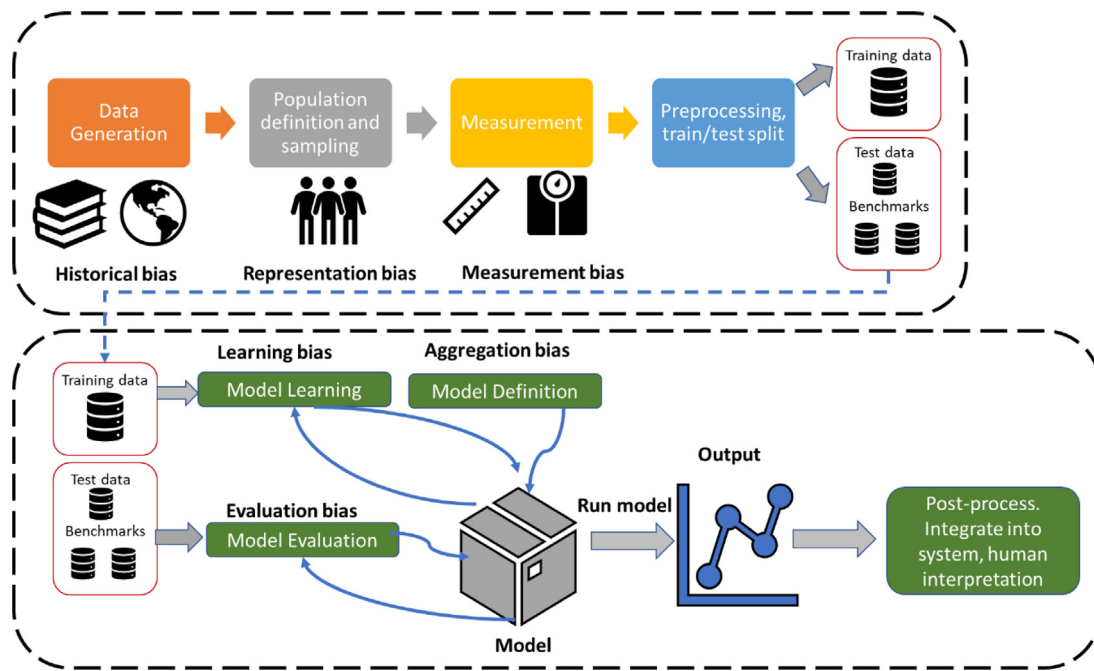


Fig. 12. A visual representation of algorithmic bias sources.

resource allocation, product quality control, or predictive maintenance decisions.

Potential bias in ML models can perpetuate and exacerbate existing inequalities. In manufacturing, for instance, this could lead to disproportionate resource allocation based on flawed data patterns or affect the accuracy of predictive models used for decision-making. To mitigate such risks, manufacturers must ensure training datasets are diverse and representative of the operational environments they are meant to model. Regular audits and evaluations of ML models, utilizing fairness metrics to test the performance over different demographic groups, are vital in addressing potential biases [289].

Lo Piano et al. [248] concluded the following about the opaqueness of many ML processes: This black-box nature in many ML systems surrenders clarity of process; therefore, when fairness and accountability are held in abeyance, it could be very hard to interrogate the reasoning behind decisions. This issue is particularly relevant in manufacturing, where transparency is crucial to avoid unintended consequences of automated decisions. Transparent models allow for better oversight and ensure ethical principles are maintained throughout the ML lifecycle.

It is worth adding that, the bias may also come from imbalanced or non-representative datasets, which can skew ML predictions [292]. Regular audits and fairness evaluations are necessary to detect such biases. Fig. 12 provides a visual representation of Sources of Algorithmic Bias, highlighting areas such as historical data and label bias, and proposes mitigation strategies to address them. One approach to combating algorithmic bias is to develop models emphasizing fairness metrics. These metrics evaluate the performance of models across different groups or conditions, ensuring that the model does not unfairly benefit or harm a specific demographic. Auditing tools and governance frameworks can ensure manufacturers regularly assess their algorithms for fairness and adjust them as necessary [293].

6.3. Accountability and transparency in decision-making

As ML systems gain autonomy in manufacturing, defining accountability becomes increasingly complex. Who is responsible for decisions made by an ML system when something goes wrong? Establishing a governance framework that outlines roles and responsibilities for developing ML frameworks is crucial.

A principal recommendation is establishing data stewards or oversight committees to manage data governance. These committees are to evaluate the deployed algorithm against the ethical guidelines of fairness, transparency, and accountability. For instance, the High-Level Expert Group on AI in the European Union proposed a set of principles for trustworthy AI that includes these considerations, calling for fairness and the protection of fundamental rights [290]. Transparency is also a key aspect of ensuring accountability. In many ML applications, especially in predictive models used in manufacturing, the “black box” nature of algorithms creates challenges for transparency and decision-making [291]. There is a growing call for more interpretable ML models where manufacturers can understand and justify the predictions or decisions made by these systems.

6.4. Incident response and ethical oversight

Ethical oversight goes hand in hand with data governance. As manufacturing increasingly relies on ML, organizations must be prepared to handle potential ethical dilemmas or data breaches. This preparation includes having an Incident Response Plan that details how to handle data breaches effectively, notify stakeholders, and comply with regulatory requirements. Moreover, as Lo Piano et al. [248] suggested, ML systems in critical industries like manufacturing require a comprehensive framework for ethical governance that integrates accountability, fairness, and transparency. Organizations should engage stakeholders, including employees, customers, and regulators, in discussions about these ethical considerations, fostering a culture of accountability and integrating diverse perspectives into decision-making.

7. A critical discussion and future perspective

Based on the above review, the manufacturing industry is clearly poised for rapid expansion in ML adaptations, while processing power and data volumes continue to grow exponentially [294]. By leveraging the powerful tools and strategies of big data analytics, it is now feasible to process high-dimensional data sets efficiently [295]. Adopting these technologies is crucial for businesses looking to remain competitive and maintain their edge in the market. In the industrial sector, ARM (association rule mining) is employed more extensively than SPM (sequential

pattern mining) [296]. However, more SPM studies, including time information during model building, could help factories react quickly to time-critical situations. This method benefits industrial organizations because analyzing common sequential patterns can offer valuable insights for predicting future events. Manufacturers can gain a competitive edge by integrating SPM with other data analytics techniques and driving their businesses forward [297].

Text mining (TM) has expanded to cover many new fields recently. However, its potential applications in manufacturing are yet to be explored [298]. This presents an exciting opportunity for future research to explore the possibilities of text mining in manufacturing, as over 80% of a company's information is in text format. Manufacturing suppliers are increasingly using text data. Techniques like association rule generation, classification, and clustering can extract insights from large data volumes. These insights can help manufacturers improve operations [299–301]. For instance, manufacturing documents can be sorted into groups or clusters based on their subject matter and structure using TM techniques. Additionally, sentiment analysis can be utilized to understand better consumer preferences and how people feel about production-related content [302].

Manufacturers today share a wealth of online information about their processes, providing valuable resource and insights that can aid similar producers in making informed decisions [303,304]. The rapid growth of online activity has spurred the development of complex digital engineering data, which can be leveraged through web mining, an automated process that collects manufacturing-related data. This data can be analyzed to extract valuable and meaningful business insights, enabling more informed decision-making [304,305]. However, the manufacturing industry is currently replete with technical jargon, and the differences between organizations can make even the most basic tasks challenging. As a result, future research in web mining for manufacturing will likely cause significant contributions to the field as manufacturers seek new ways to extract actionable insights from the vast amounts of data available.

Several ontology-based solutions for managing production records have been implemented to provide a semantic representation of information in recent years. However, it is anticipated that more research will be conducted over the next few years to develop ontology-based manufacturing systems that incorporate ML methodologies. Ontology-based schemes are a reliable means for eliminating semantic ambiguity, leading to the creation of more precise and trustworthy decision-support tools [306,307]. This necessitates the development of new machine-learning techniques. It is worth emphasizing that there is no universal solution for ontology in the manufacturing sector, and multiple ontologies can coexist. By leveraging the capabilities of ontology-based systems in conjunction with other data analytics approaches, manufacturers can obtain more profound insights into their production processes, make better-informed decisions, and propel their businesses toward success [308,309].

Supervised and unsupervised learning have been the focus of more attention than RL in manufacturing engineering [310]. However, RL provides a robust framework for analyzing and testing systems based on rewards and penalties. It can autonomously complete tasks without external guidance, allowing to adapt to new and challenging situations. Reinforcement learning is particularly valuable for tackling complex decision-making tasks and addressing planning and control challenges in the manufacturing industry. As the potential benefits of RL become increasingly evident, it is expected that the engineering sector will increasingly prioritize this powerful technique in the near future [311, 312]. The k-means algorithm, among unsupervised machine learning algorithms, has been very commonly used in clustering for manufacturing research. However, future researchers could explore refining the k-means++ method to enhance its effectiveness and accuracy. It is worth noting that relatively little research has been done on k-means++ in the manufacturing industry [51]. On the other hand, DBSCAN is

an alternative clustering method with great potential. It is particularly well-suited to handling noisy data and can be highly adaptive in forming clusters. As such, it may achieve greater prominence in manufacturing research in the coming years [313].

Data processing can be performed in batch or streaming mode, depending on the requirements [295]. In real-time processing, collected data is updated with the model as new information is gathered. Some industries employ batch data processing, a technique whereby several interactions are accumulated over time, processed, and then used to create a model. However, data can also be processed in real-time or stream processing, where data is processed continuously as it is generated. Despite the potential benefits of stream processing in the manufacturing industry, most research has focused on batch processing. Therefore, it is imperative to direct future research toward developing manufacturing applications for real-time data extraction and ML, as it is an essential and demanding field [314–316].

Additionally, federated learning and edge computing remain largely unexplored in manufacturing. As industrial environments generate increasingly large datasets, centralized cloud-based ML solutions may struggle with latency and security concerns [315–324]. Implementing decentralized ML models through edge computing can enable real-time analytics while preserving data privacy. Research into these technologies will be critical for scalable and secure ML-driven manufacturing. The manufacturing industry is increasingly embracing the use of robots in its operations. Industrial robots are already employed in numerous factories worldwide and are receiving widespread attention from the media. It is expected that soon, ML-based applications will become increasingly vital in manufacturing, alongside the emergence of intelligent workshops and the expanded use of manufacturing robots [319,320].

8. Conclusion

This review first presented a broad perspective on how ML is reshaping manufacturing sectors by enhancing productivity, predictive capabilities, and system responsiveness. Predictive maintenance is currently among the prominent target applications, reducing unplanned downtime and optimizing maintenance schedules [266,321]. Deep learning models, especially CNNs, have significantly improved visual quality inspection tasks by automating complex pattern recognition [182,262,322]. However, challenges related to training data, such as data imbalance, noise, and annotation costs, still limit the scalability of AI models in the field. The review highlighted alternative methods like data synthesis, transfer learning, and semi-supervised techniques that can help address these limitations [22,228,323,324].

While the open literature clearly reinforces the commonly accepted view that “ML adoption is fast accelerating in different manufacturing sectors”, critical issues like model transparency, data quality, and operational scalability remain largely unresolved [325–338]. XAI techniques, such as SHAP and LIME, are deemed critical for future smart manufacturing, addressing one of the most persistent barriers in industrial AI: interpretability. Explainable AI models are key for performance, compliance, and operator trust in high-risk, regulated environments like aerospace or pharmaceuticals. XAI should not be considered a peripheral concern in ML adoption, but a central requirement in modern manufacturing systems. Furthermore, integrating ML into industrial platforms like MES and IIoT remains an evolving frontier. Edge computing has emerged as a key enabler, allowing real-time model inference close to the data source, thus reducing latency and enhancing responsiveness [257,258].

Ethical considerations equally play a crucial role in efficient ML implementations in manufacturing. The growing reliance on automated decision-making tools can bring forward fairness, accountability, and bias concerns. Manufacturing firms need clear policies and frameworks that guide the responsible use of ML, particularly when decisions affect

the product quality, user and operator safety, and regulatory compliance. In the same relation, combining e.g. Quality Function Deployment (QFD) approaches [339] with machine learning (ML) in manufacturing can assist in adapting new user/operator-driven decisions that are data-informed, adaptive, and scalable. Moreover, continual workforce development is vital.

Finally, it is worth noting that many small-to-medium sized organizations still struggle to extract real value from AI initiatives without investing in reskilling programs. More systematic collaborations between academia, industry, and regulatory bodies are deemed essential in this regard, to co-develop industry-relevant training tools embodied with a culture of ethical innovation.

CRedit authorship contribution statement

Yassmin Seid Ahmed: Conceptualization, Methodology, Writing – original draft. **Abbas S. Milani:** Writing – review & editing, Project administration.

Ethical statement

The review paper is original, has not been submitted to another journal, and follows ethical considerations. Sources have been appropriately acknowledged, and there is no plagiarism, data fabrication, or improper reuse of material.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available upon request.

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